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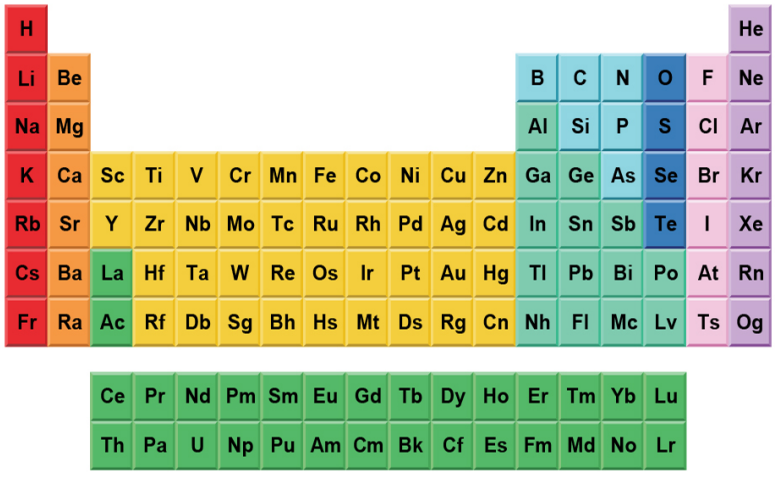
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Review on Recycling of Microplastics in Cigarette Butts

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Abstract. It's a lesser-known fact that the cigarette filter is made of cellulose acetate, a kind of non-biodegradable plastic and it takes almost 10 years for this ingredient to completely decompose. Cigarette smoking is on the rise throughout the world. Ever since tobacco smoking began there are both opponents and proponents for the same. Instead of getting into a discussion as to whether one should smoke or not; would like to focus on cigarette butts which a micro plastic that is harmful to the environmentand needs to be disposed of in an appropriate manner to conserve the environment and human health. As the trend is today, cigarette smoking is on the increase. And in light of this event recycling cigarette butts give rise to a sustainable material.
This paper will cover how cigarette butts can be collected and recycled to tackle the unfriendly and unsightly remains of cigarettes after smoking.
Keywords: Cigarette butts, Solutions to handle cigarette butt waste, Awareness regarding probable health and environmental hazards

1. Introduction

Filters were initially used from 1860 to 1920 to prevent particles of tobacco from entering the mouth. Inthe 1950s after published studies began to reveal the health threats of tobacco, the cigarette industry propagated the use of cigarettes with filters. The use of filters gave cigarette smokers an illusion that they are safe from the ill effects of cigarette smoking and started inhaling as muchor more tar, nicotine, and noxious gases as smokers of unfiltered cigarettes. The fast life of cities and the increased pressure in day-to-day life due to the industrial revolution causing men to competewith machines has caused many men to smoke to relieve stress and tension. Consequently, there is an increased litter of cigarette butts.

2. In what way are cigarette butts harmful?

2.1. What are micro plastics?

Micro plastics comprise tiny fibers from nylon clothes and other synthetic textiles. Plastic particles smaller than 5 millimeters are considered to be micro plastics. Well, it is likely that microplastics continue breaking down even past 10nm. Plastics do not blend with the soil easily but remain in the soil in a particulate manner, probably just becoming into even smaller pieces. The tinier the particle, the easier it is for them to pass cell membranes and cause tissue damage. For example, it is observed that micro plastics can absorb toxic chemicals and then release them into an animal's digestive system. This would definitely be harmful to our health. [1]
Research has shown these nano plastics can enter the intestines during digestion. Whether they can intermingle with the blood is not clear, however, another rather unnerving study demonstrated thatnano plastic particles lodged in the brains of fish affected their behavior. The study found that

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plastic particles made fish eat slowly and more lethargic. Research has not yet established Nano plastics penetrate brain tissue in humans, let alone affect behavior. The filter of cigarette is made of micro plastic. Plastics cannot be digested or decomposed and once produced, one cannotget rid of them. They are mistaken for food by animals and ingested blocking the digestive systemof animals due to which there are low levels of oxygen and consequently reduced energy levels inanimals. Very tiny micro plastics get embedded in the animal tissues. They enter human beings through the food chain. Micro plastics find their way to humans through ingestion or respiration.

2.2 What are micro plastics in cigarettes made up of?

The majority of cigarette filters are made of cellulose acetate, which is a type of plastic. The cigarette filter is not cotton, but a plastic that does not disintegrate similar to other types of plasticin the environment. Cigarette butts (CB) are one of the most common types of litter in cities, withan estimated 4.5 trillion thrown each year, accounting for22-46 percent of visible litter. 76 percentof cigarettes smoked in public in metropolitan areas are littered rather than disposed of properly. Butts are then carried to coastal locations by rain or river water, where they are typically the mostabundant part of litter collected. The chemical makeup of cigarette butts waste has an impact on the environment. Because cigarette butts are made up of a highly twisted network of fibers combined with plasticizers, the breakdown is sluggish and difficult. Furthermore, toxic metals and nicotine have been found to seep out of cigarette butts. The Lethal Concentration 50 for test fish exposed to cigarette butts' leachates is one cigarette butt immersed in a liter of water for 96 hours.When this trash is released into the environment, it poses a significant risk to living species and ecosystem health. As a result, cigarette butts pose a threat to the environment and public health. [2]

3. Where are cigarette butts found?

Cigarette butts are tossed on the street or littered in parks, and beaches, they may be thrown in street gutters too. Since indoor smoking is banned by many organizations smokers have been pushed outdoors. Cigarette butts are mostly found near corporate offices, hospitals, colleges, and tourist destinations. Also, outside bars cigarette butts are found in large quantities.

4. How can these cigarette butts be effectively collected?

Countries are beginning to put the burden of cleaning up these butts onto cigarette companies themselves. People who collect cigarette butts for the purpose of recycling are termed as heroes because common men due to lack of education and ignorance do not like to endeavor in this field for prestige sake. One of the ways of collecting cigarette butts is to provide special fire-safe receptacles. It would be wise to charge a percentage of the cost of the cigarettepack as a clean-up cost. Attractive crucibles could be placed on the street to promote the appropriate disposal of cigarette butts, as shown in Figures 1a and 1b.



Figure 1. a) California b) Seattle

At present companies who recycle cigarette butts in India pay a collection fee of Rs. 700 for everykilogram and Rs. 80 for every 100 grams of cigarette waste.

5. What is harmful about cigarette butts?

Cigarette butts take anywhere from 18 months to ten years to decompose.
Chemicals and heavy metals, the same things that kill smokers, leak into the environment whenbutts are discarded. They frequently wash into the gutters, polluting our rivers and endangering fish and other creatures.
The most serious issue is the filters, which are mostly comprised of cellulose acetate, a type ofplastic. That means the filters take a long time to biodegrade, even when they do, thousands of tiny plastic fibers wind up in the environment, streams, and wildlife. Arsenic, benzene, and lead are among the substances found in the fibers.
Cigarette butts are hazardous plastic fragments that are frequently mishandled and contain over15,000 separate strands of plastic fiber. Cigarette butts can be carried into rivers and lakes, eventually ending up in the ocean. This source might potentially release 300,000 tons of micro-plastic fibers into the aquatic ecosystem each year.
Furthermore, harmful compounds such as nicotine, carcinogenic tar, and polycyclic aromatichydrocarbons have a highly toxic effect on aquatic creatures, causing serious damage.
According to the WHO, tobacco product waste contains over 7,000 toxic chemicals. Cellulose acetate, the major component of cigarette butts, along with paper and rayon, pollutes land and water. A report by Truth Initiative, a U.S.-based NGO says that the chemicals that leach from a single cigarettebutt can release enough toxins to kill 50% of saltwater and freshwater fish exposed to it for 96 hours. [3,4]
There is also the fear of birds, fish, and animals eating cigarette butts that can be disastrous," viii. As a result, it is vital to comprehend the mechanism by which cigarette butts release micro Plastic fibers, the potential environmental impact, and viable mitigation techniques for cigarette butt litter. [5]

6. How is the cigarette butt processed for recycling?

The cigarettes get shredded and separated into 2 key parts, the organic matter which is the ash, tobacco, and paper composted and the inorganic component which is the filter is melted.
The paper is ground in an industrial grinder and treated with an organic binder. The ground material is spread into sheets and dried. They are later cut and sold in local shops as mosquito repellant. The paper with remains of nicotine act as a mosquito repellant. To use the mosquito repellant, it just has to be burnt.
The plastic in the cigarette butt is also ground in the industrial grinder, they are then soakedin sterilizing chemicals for 24 hours which leaves them looking like cotton. Later the plastic fibersare fluffed and made to the required specifications.
It is not necessary that the plastic from cigarette butts be recycled separately. It can be combined with other plastics such as polyethylene and polypropylene. Cellulose acetate is found in objects like combs, toothbrushes, playing cards, tool handles, pen barrels, etc. made from plastic. Hence all these plastics can be recycled together. [6]

7. What can be made from cigarette butts?

- Composted organic waste from cigarette butts is used as fertilizers.
- At present in India cigarette butts are recycled and used to make mosquito repellants and soft toys.
- Butts is also used to make packing and building material as well as plastics for key chainsand toys and cushion. [7]
- In France the fibers from cigarette butts are used to make jackets, here the plastics obtained from cigarette butts can be used as an insulating material, as shown in figure 2 a & 2 b. [8]

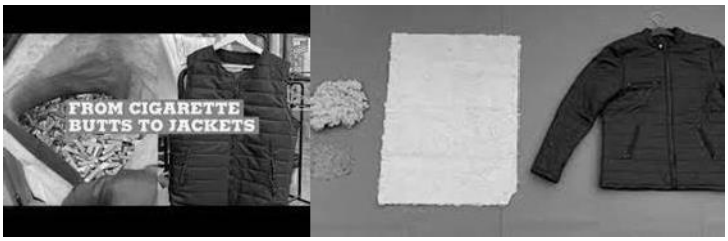


Figure 2. Insulating material from CB used for jackets in France

If the cigarette butt is not recycled into an industrial or commercial product it is converted into pellets which can be used in any plastic industry Figure 4.



Figure 3. Pellets of plastic from cigarette butts

Also, France manufactures benches and furniture from plastic obtained from cigarette butts figure 5.



Figure 4. Benches made from CB in France

8. How cigarette butts can be used in the construction industry.

Cigarette butts with leftover tobacco had an energy value of 16.5 MJ/kg. If just 2.5% of all bricks produced annually worldwide included 1% CB content, all of the CBs currently produced could be recycled in bricks, and it is estimated that global firing energy consumptioncould be reduced by approximately 20 billion MJ (mega joules). This is roughly equal to the annual energy consumption of one million houses in Victoria, Australia. “Firing butts into bricks is a reliable and practical way to deal with this terrible environmental problem, whileat the same time cutting brickmaking production costs,” Mohajerani said.

Cigarette butt bricks are also lighter and provide superior insulation, resulting in lower heating and cooling expenditures in the home. Mohajerani’s research team discovered that once cigarette butts were employed on the pavement, they could not only withstand high traffic, butit could also minimize heat created by the sun hammering down on black asphalt, aphenomenon known as the urban heat-island effect. It was discovered that butts encapsulatedin bitumen and combined with class 170 bitumen performed better in heavy traffic than paraffin-encapsulated butts mixed with c170 and c320 bitumen after completing various testson the samples. The butts encased in paraffin were barely able to withstand light traffic. [9]

Insulating material that can be used in the construction industry can be made from the plastic in cigarette butts.

9. Is recycling cigarette butts a small business?

About 6 trillion cigarettes are produced every year, leading to more than 1.2 million tons of cigarette butt waste. These figures are expected to increase by more than 50 percent by 2025,mainly due to an increase in the world population.

It is estimated that 4.5 trillion cigarette butts are disposed of annually. An estimated 766,571metric tons of cigarette butts make their way into the environment every year.

10. Has the litter of cigarette butts reduced wherever an attempt has been made?

By the financial year 2019-2020 a whopping 10,000 kg of cigarette waste, of which 40 percent has been recycled and utilized. Each kg of cigarette waste comprises 4,000 cigarette butts. A company by the name CODE has pioneered in India recycling cigarette butts. Having started in 2016 in Noida so far the duo, Vishal Kanet and Noman Gupta have successfully collected 500-600 kilograms of cigarette waste and recycled 40-50 kgs. CODE stands for Conservation of the depleting environment.

11. Who should give a go-ahead to the project?

Non-smokers should give a go-ahead for this method of recycling micro plastics. When smokers are careless about the harm they do to their own bodies by smoking, non-smokers who strongly oppose this habit should propagate the ill effects of the micro plastic environment and come up with innovative methods of freeing the environment of this nuisance.

12. Scope of this innovative method

Before anything can be done technically to recycle cigarette butts, education has to play a major rolein this field. Non-smokers who understand the harm of cigarettes should take the lead in recycling cigarette butts. Social stigma should not be feared and a bold and confident approach should be taken to recycling this notorious element. Smokers should be educated about the harm of cigarette butts and should be encouraged to dispose of cigarette butts in the proper manner. The misconception that cigarettebutts are made of cotton which is biodegradable should be changed and people need to know that cigarette butts are ma of non-biodegradable micro plastic. Fines may be imposed on those who arecareless in this matter. Government should finance start-ups in this area and motivate them towards a cleanIndia and world.

13. Conclusion

Like every drop makes an ocean, every cigarette butt adds up to the litter which amounts to 4.5 trillion individual butts polluting our global environment. Due to a long history of the tobacco industry targeting, low-income communities of color, they have greater smoking rates and confront systemic disparities that make it difficult for them to access health information and resources. Furthermore, they are disproportionately served by inequitable waste management, which can exacerbate the consequencesof cigarette butt pollution. Hence recycling cigarette butts makes a sustainable innovative solutionto the problem.

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